

The **BLE-Based Employee Tracking System** aims to automate the attendance and work hour monitoring process for employees using BLE technology. The system provides a scalable, touchless, and low-maintenance alternative to traditional methods such as RFID cards and biometric systems.

Key Functionalities Covered in This Project:

1. BLE Beacon-Based Attendance

Employees are assigned BLE beacons that continuously transmit unique UUID signals. These beacons serve as identity tags for automatic detection.

2. ESP32-Based BLE Receiver

The ESP32 microcontroller detects beacon signals within a defined range (approximately 30 meters) and logs the entry and exit times for each employee.

3. Wi-Fi Connectivity and Firebase Integration

The ESP32 uploads the captured data to a cloud-based backend using Wi-Fi. Firebase is used to store, manage, and process real-time attendance data.

4. Work Hour Calculation

The system calculates the total active time of each employee based on their daily entry and exit logs.

5. Web Dashboard for Monitoring

A user-friendly dashboard is developed using web technologies to allow remote access to employee logs, time summaries, and status updates.

6. Multi-Device Accessibility

The system can be accessed from desktops, laptops, or mobile browsers, enabling flexibility and convenience for site supervisors or HR managers.

Out of Scope:

- Integration with payroll processing systems
- Mobile app development (only web interface is considered)
- BLE beacon battery optimization or power-saving algorithms
- Indoor positioning or real-time location tracking within a site

This system is designed to be **easily expandable**, allowing future upgrades such as notification systems, app-based controls, location tracking, and integration with external attendance or payroll software. It lays a strong foundation for building an end-to-end workforce automation platform for smart industries and digital workplaces.